

Upper bounds on the lattice gap: the Feynman–Bijl inequality, the abelian structure factor and the non-abelian dressing obstruction

The single-mode (Feynman–Bijl) inequality bounds the gap of the Kogut–Susskind Hamiltonian above by ground-state expectation values, and three consequences follow with explicit constants.

(i) For any real gauge-invariant function O of the links, $\Delta_{a,L}^{\text{phys}} \leq \frac{\hbar c g^2}{2a} \sum_{\ell} \langle |\nabla_{\ell} O|^2 \rangle / \text{Var } O$; for a plaquette this is $\Delta \leq \hbar c g^2 N / (a \text{Var Re tr } U_p)$, which together with the strong-coupling lower bound shows that for large g the gap is of order $g^2 \hbar c / a$ on both sides. (ii) For compact $U(1)$ the operator $O = \sum_{\ell} \varphi_{\ell} E_{\ell}$ is gauge invariant without dressing, and the uncertainty principle turns the inequality into

$$\Delta \leq \frac{2\hbar c}{a g^2} \frac{\hat{S}(k)}{\langle \cos \theta_p \rangle}, \quad \hat{S}(k) = \frac{\langle \Theta_k^2 \rangle}{\sum_p |(d\varphi_k)_p|^2}, \quad \Theta_k = \sum_p (d\varphi_k)_p \sin \theta_p,$$

for every lattice momentum k : the abelian gap is bounded by the zero-momentum limit of the plaquette-sine structure factor, so a Coulomb phase, where $\hat{S}(k) \rightarrow 0$, is gapless, and T2' fails for $U(1)$ exactly through this channel. (iii) For a non-abelian group no operator linear in the electric field with c-number coefficients commutes with Gauss's law (proved), so the same trial excitation must be dressed with a Wilson line, and the dressing contributes to the double commutator an electric term of order $g^2 \hbar c / a$ per unit length that does not vanish as the wavelength grows: the channel through which the abelian gap closes is blocked by gauge invariance, which is the exact place where the non-abelian conjecture T2' parts from the abelian theorem. The plaquette bound diverges in the continuum limit because plaquette fluctuations are ultraviolet dominated, so the finiteness half $m < \infty$ of the conjecture needs smeared operators, and the next target is the same inequality for Wilson-flowed observables with a flow-Jacobian bound uniform in a . Nothing here is promoted.

1. The inequality

Let H be self-adjoint and bounded below with simple ground state ψ_0 and next eigenvalue E_1 , and let O be symmetric with $\psi_0 \in D(O)$, $O\psi_0 \in D(H)$ and $\langle \psi_0, O\psi_0 \rangle = 0$. Then $O\psi_0 \perp \psi_0$, so by the min-max principle

$$E_1 - E_0 \leq \frac{\langle O\psi_0, (H - E_0)O\psi_0 \rangle}{\|O\psi_0\|^2} = \frac{\frac{1}{2} \langle \psi_0, [O, [H, O]]\psi_0 \rangle}{\langle \psi_0, O^2\psi_0 \rangle},$$

using $[O, [H, O]] = 2OHO - O^2H - HO^2$ and $(H - E_0)\psi_0 = 0$. If O commutes with the Gauss generators, $O\psi_0$ lies in the physical sector and the same bound holds for Δ^{phys} . (Feynman, Phys. Rev. 94 (1954) 262, metadata; the argument is the two lines above.)

2. Functions of the links, and the strong-coupling order

Take $H = \frac{\hbar c}{a} [\frac{g^2}{2} \sum_{\ell} (-\Delta_{\ell}) + \frac{2}{g^2} \sum_p (N - \text{Re tr } U_p)]$ as in the obligations map §1 and a real function $O = f(U)$ with $\langle f \rangle = 0$. Then $[V, f] = 0$, and for the Laplacian $\Delta_{\ell} = \sum_a X_a^2$ in the left-invariant fields X_a of link ℓ ,

$$[f, [-\Delta_{\ell}, f]] = 2 \sum_a (X_a f)^2 =: 2|\nabla_{\ell} f|^2,$$

because $[\Delta, f]h = (\Delta f)h + 2 \sum_a (X_a f)(X_a h)$ and the second commutator with f removes the (Δf) terms and leaves $-2 \sum_a (X_a f)^2 h$. Hence

Proposition 1. For every real gauge-invariant $f(U)$ with $\langle f \rangle_0 = 0$,

$$\Delta_{a,L}^{\text{phys}} \leq \frac{\hbar c g^2}{2a} \frac{\sum_{\ell} \langle |\nabla_{\ell} f|^2 \rangle_0}{\langle f^2 \rangle_0}.$$

Corollary 2 (plaquette). With $f = \text{Re tr } U_p - \langle \text{Re tr } U_p \rangle_0$: for a link ℓ of p , $X_a \text{tr } U_p = \text{tr}(T^a M)$ up to a phase, with M a cyclic rearrangement of U_p , so by the $SU(N)$ completeness relation $\sum_a \text{tr}(T^a M) \text{tr}(T^a M^\dagger) = \frac{1}{2}[\text{tr } M M^\dagger - \frac{1}{N}|\text{tr } M|^2] \leq \frac{N}{2}$, and $|\nabla_\ell \text{Re tr } U_p|^2 \leq |\nabla_\ell \text{tr } U_p|^2 \leq N/2$. Four links give $\sum_\ell |\nabla_\ell f|^2 \leq 2N$ and

$$\Delta_{a,L}^{\text{phys}} \leq \frac{\hbar c g^2 N}{a \text{Var}_0(\text{Re tr } U_p)}.$$

Corollary 3 (two-sided strong-coupling order). At $g = \infty$ the ground state is the Haar measure and $\text{Var}(\text{Re tr } U_p) = \frac{1}{2}(\langle |\text{tr } U|^2 \rangle + \frac{1}{2} \text{Re} \langle (\text{tr } U)^2 \rangle)$ equals 1 for $SU(2)$ and $\frac{1}{2}$ for $SU(N \geq 3)$, by Peter–Weyl orthogonality of the fundamental character and the vanishing of $\int (\text{tr } U)^2 dU$ for $N \geq 3$. The ground-state expectation of a local function is continuous in g^{-4} at $g = \infty$ uniformly in the volume (Yarotsky’s Theorem 1, conclusion 4, analyticity of $\omega(A)$ in the perturbation parameter; the finite-volume statement is analytic perturbation theory of a simple isolated eigenvalue), so there is $g_1(N)$ with $\text{Var}_0 \geq \frac{1}{4}$ for $g \geq g_1$. Combined with the strong-coupling note,

$$\gamma \frac{g^2}{2} \frac{N^2 - 1}{2N} \frac{\hbar c}{a} \leq \Delta_{a,L}^{\text{phys}} \leq 4N g^2 \frac{\hbar c}{a} \quad (g \geq \max(g_0, g_1)),$$

uniformly in the lattice size. The strong-coupling gap is $g^2 \hbar c/a$ times a number bounded on both sides.

Why the plaquette says nothing about the continuum. As $a \rightarrow 0$ along the scaling curve the links approach the identity and $\text{Var}(\text{Re tr } U_p) \rightarrow 0$ (the plaquette angle fluctuates on the scale g , and its square on the scale g^4), so the right side of Corollary 2 grows without bound. The plaquette is an ultraviolet observable whose excitation costs an energy of order $\hbar c/a$; the finiteness half $m < \infty$ of the conjecture requires an operator whose ratio $\sum_\ell \langle |\nabla_\ell O|^2 \rangle / \text{Var } O$ stays of order a/ℓ for a physical length ℓ . Section 5 states that target.

3. The abelian theory: the gap is bounded by a structure factor

For compact $U(1)$ write $U_\ell = e^{i\theta_\ell}$, $E_\ell = -i\partial/\partial\theta_\ell$, $\theta_p = \sum_{\ell \in \partial p} \epsilon_{p\ell} \theta_\ell$, and

$$H = \frac{\hbar c}{a} \left[\frac{g^2}{2} \sum_\ell E_\ell^2 + \frac{1}{g^2} \sum_p (1 - \cos \theta_p) \right]$$

(other conventions rescale g^2). Gauss’s law generators $G_x = \sum_{\ell \ni x} \pm E_\ell$ commute with every E_ℓ , so

Lemma 4. $O_\varphi = \sum_\ell \varphi_\ell E_\ell$ with real φ is gauge invariant, and $\langle O_\varphi \rangle_0 = 0$.

Proof. The E_ℓ commute among themselves. Charge conjugation $\theta \mapsto -\theta$ is a unitary symmetry of H under which the unique ground state is invariant and $E_\ell \mapsto -E_\ell$. $\square$

With $[E_\ell, h(\theta)] = -i\partial_\ell h$ and the lattice curl $(d\varphi)_p = \sum_{\ell \in \partial p} \epsilon_{p\ell} \varphi_\ell$: $[O_\varphi, \cos \theta_p] = i(d\varphi)_p \sin \theta_p$, $[O_\varphi, \sin \theta_p] = -i(d\varphi)_p \cos \theta_p$, hence

$$[O_\varphi, [H, O_\varphi]] = \frac{\hbar c}{ag^2} \sum_p (d\varphi)_p^2 \cos \theta_p,$$

and Section 1 gives

$$\Delta^{\text{phys}} \leq \frac{\hbar c}{2ag^2} \frac{\sum_p (d\varphi)_p^2 c_0}{\langle O_\varphi^2 \rangle_0}, \quad c_0 = \langle \cos \theta_p \rangle_0,$$

where c_0 is independent of p by translation invariance of the unique ground state. To bound $\langle O_\varphi^2 \rangle$ below use the uncertainty relation with the gauge-invariant $\Theta = \sum_p \psi_p \sin \theta_p$: $\langle O^2 \rangle \langle \Theta^2 \rangle \geq \frac{1}{4} |\langle [O, \Theta] \rangle|^2$ with $\langle [O_\varphi, \Theta] \rangle = -ic_0 \sum_p \psi_p (d\varphi)_p$. Choosing $\psi = d\varphi$:

Theorem 5. If $c_0 > 0$, then for every real φ on the links,

$$\Delta_{a,L}^{\text{phys}} \leq \frac{2\hbar c}{a g^2 c_0} \frac{\langle \Theta_\varphi^2 \rangle_0}{\sum_p (d\varphi)_p^2}, \quad \Theta_\varphi = \sum_p (d\varphi)_p \sin \theta_p.$$

For a plane wave $\varphi_\ell = \cos(k \cdot x_\ell) u_{i(\ell)}$ the ratio is the structure factor $\hat{S}(k)$ of the plaquette sine at momentum k , and $\Delta^{\text{phys}} \leq 2\hbar c \hat{S}(k)/(a g^2 c_0)$ for every $k \neq 0$ allowed by the box.

Proof. Insert the uncertainty bound into the previous display: $\langle O^2 \rangle \geq c_0^2 (\sum_p (d\varphi)_p^2)^2 / (4 \langle \Theta^2 \rangle)$.

□

Reading. In a Coulomb phase $\sin \theta_p$ is a curl of a nearly Gaussian field and $\hat{S}(k) \rightarrow 0$ as $k \rightarrow 0$; the theorem then forces $\Delta \rightarrow 0$ with the box, which is the gaplessness Guth and Fröhlich–Spencer prove at weak coupling (abstract, B78), obtained here as a consequence of two ground-state quantities, c_0 and $\hat{S}$. In a confining phase $\hat{S}(0) > 0$ and the bound is finite, consistent with the strong-coupling gap. The theorem does not decide which phase occurs; it shows that the abelian question T2' is equivalent, at the level of upper bounds, to the behaviour of one correlation function at zero momentum.

4. The non-abelian obstruction

Proposition 6. Let G be non-abelian and $O = \sum_\ell \sum_a \varphi_\ell^a E_\ell^a$ with c-number coefficients. If O commutes with all Gauss generators, then $\varphi = 0$.

Proof. Write $E_\ell^a = L_\ell^a$ for the left generators of link ℓ , which satisfy $[L_\ell^b, L_\ell^a] = i f^{bac} L_\ell^c$, and let R_ℓ^b be the right generators, which commute with all L_ℓ^a . The Gauss generator at x is $G_x^b = \sum_{s(\ell)=x} L_\ell^b + \sum_{t(\ell)=x} R_\ell^b$ up to the sign convention. Then $[G_x^b, O] = i \sum_{s(\ell)=x} f^{bac} \varphi_\ell^a L_\ell^c$, with no contribution from the entering links. The L_ℓ^c are linearly independent operators, so $[G_x^b, O] = 0$ for all b, x forces $f^{bac} \varphi_\ell^a = 0$ for all b, c and every link, that is $[\varphi_\ell, T^b] = 0$ for all b : φ_ℓ lies in the centre of the Lie algebra, which is trivial for a semisimple G . □

The trial excitation of Theorem 5 therefore does not exist in the non-abelian theory. A gauge-invariant operator linear in E must carry a Wilson line, $O = \sum_\ell \varphi_\ell \text{tr}(E_\ell W_{C_\ell})$ with C_ℓ a closed path through ℓ , and then $[K, O] \neq 0$: the double commutator acquires the term $\frac{\hbar c g^2}{2a} \sum_{\ell' \in C} \langle |\nabla_{\ell'} O|^2 \rangle$ of Section 2, of order $g^2 \hbar c / a$ per link of C , which does not decrease when φ_ℓ becomes a long-wavelength mode. The Feynman–Bijl bound then reads

$$\Delta^{\text{phys}} \leq \frac{\text{magnetic term } (\propto g^{-2} \hat{S}\text{-like}) + \text{string term } (\propto g^2 |C|)}{\langle O^2 \rangle},$$

and no choice of φ removes the second numerator. This is the precise sense in which gauge invariance blocks, for non-abelian groups, the channel through which the abelian gap closes: the excitation that would be a photon must be attached to a string whose electric energy is set by the cutoff-scale coupling. It is an obstruction to proving gaplessness, and it is consistent with T2'; it proves nothing in the direction of a lower bound.

5. The finiteness half and the flowed-operator target

Jaffe–Witten require $m < \infty$. On the lattice every bound above is finite, and the continuum question is whether some gauge-invariant observable has $\frac{\hbar c g^2}{2a} \sum_\ell \langle |\nabla_\ell O|^2 \rangle / \text{Var } O$ bounded as $a \rightarrow 0$ along the scaling curve. Plaquettes fail (Section 2). Wilson-flowed observables (Lüscher's flow, local companion, equations (1.1)–(2.4) at passage level) are smooth functions of the links smeared over the physical radius $\sqrt{8t}$; for them $\nabla_\ell O_t$ is the flow Jacobian, and the target statement is

T_{fin}. For the flowed energy density $O_t = t^2 \text{tr } F_t^2$ at fixed physical flow time t , $\sum_\ell \langle |\nabla_\ell O_t|^2 \rangle$ is $O(a^3 t^{-3/2})$ times a number uniform in a , so that $\Delta \leq C(t) \hbar c / \sqrt{8t}$ in the continuum limit.

The ingredient to prove is a diamagnetic-type bound on the linearized flow, which is a covariant heat equation with a curvature term; the covariant heat kernel obeys the scalar Gaussian bound by Kato's inequality, and the curvature term must be controlled by the flow's own a-priori estimates. This is the next theorem-sized target on the upper side; it delivers the finiteness half of the conjecture in the finite-volume continuum theory once existence is available, and its proof would be the first continuum-uniform spectral statement in this programme.

6. Consequence for STATE

The upper side of the lattice route is organized: Proposition 1 is the tool, Corollary 3 fixes the strong-coupling gap to order $g^2\hbar c/a$ on both sides, Theorem 5 reduces the abelian T2' to a structure factor, and Proposition 6 with the string term is where the non-abelian theory escapes that reduction. Two targets follow: T_{fin} via flowed observables (upper side, continuum), and, on the lower side, the weak-coupling lattice bound of STATE step 3 whose obstruction is now sharper: any lower-bound argument must use gauge invariance in the way Proposition 6 shows the upper bound is forced to.